Certified Transcription Triage: a Distribution-Free Accept/Defer Gate with an Audited Confidence Signal for Mandarin ASR

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***Abstract*—Automatic speech recognition (ASR) is increasingly deployed with a human in the loop, auto-accepting confident transcriptions and deferring the rest. This raises two questions: can we certify, distribution-free, that the character error rate (CER) of the auto-accepted set stays below a target, and do field-standard confidence scores beat honest random deferral? We build `asr-gate`, which (i) issues a Learn-then-Test selective-risk certificate on the accepted-set macro-CER and (ii) audits each score's excess area-under-the-risk–coverage-curve against an analytic random-deferral baseline under a matched-abstention permutation null with Holm control. On Aishell-1 (Paraformer) the certificate holds at $\alpha = 2\%$, $\delta = 0.1$ with 0/20 violations across 20 correlated reseeds of one fixed test set, at 85.7–96.2% acceptance and accepted-set macro-CER 1.00–1.53% (full-set 1.98%); the binding sub-base-rate target is $\alpha = 1.9\%$ ($\alpha = 1.5\%$ is calibration-budget-conditional). A frozen backbone$\times$corpus landscape (three architectures $\times$ three Mandarin corpora) breaks the single-cell reading: a documented-open-data backbone (Belle-whisper, a different architecture and organization, training data excluding the test) also certifies non-vacuously on Aishell-1 test ($\alpha = 5\%$, 0/20), bounding the undisclosed-training-data contamination concern by independence; a fully-open transducer failed to expose usable posteriors and is reported non-certifying. Weak Whisper and English (LibriSpeech) arms are honestly vacuous at tight targets. In the audit, all twelve cells of a roster-derived family show positive excess area (0.012–0.051) with speaker-blocked bootstrap 95% CIs excluding zero, and named confidence baselines coincide with our scores — the contribution is the certificate, not a new score. All numbers reproduce from frozen result files.**

***Index Terms*—Speech recognition, selective prediction, risk control, conformal prediction, error-rate certification, distribution-free calibration.**

I. Introduction

A Practical ASR deployment rarely needs every utterance transcribed automatically; it needs the accepted utterances to be reliable and the rest routed to a human. This is a selective-prediction problem on a sequence output — the modern instance of the classical error/reject tradeoff [1], whose risk-coverage curve was formalized for learned classifiers by El-Yaniv and Wiener [2] and later integrated end-to-end into deep networks [3], [4], [5] — and it raises a guarantee question the field-standard confidence pipeline does not answer: if we auto-accept every utterance whose confidence exceeds a threshold, what can we promise about the error rate of what we accepted? A promise that is *distribution-free* — valid without modeling the error distribution — is exactly what conformal risk control provides in other settings [6], [7], [8].

We make two claims and support each with an independent body of evidence.

(C1) Certified selective triage. `asr-gate` certifies, at preregistered target α on CER and confidence level $1 - \delta$, that the auto-accepted set's macro-CER is at most α, at non-trivial coverage on Aishell-1 test, with an empirical violation rate over 20 calibration reseeds at most δ. The estimand — selective risk — is *non-monotone* in the threshold, so plain conformal risk control for monotone losses [6] does not apply; we use Learn-then-Test [9] on a reformulated bounded-mean statistic (section II-B).

(C2) A debunk-or-confirm audit. Field-standard ASR confidence scores [10] either do or do not beat honest random deferral. We test each score's excess area-under-the-risk-coverage-curve (excess-AURC) against an analytic random-deferral baseline under a matched-abstention permutation null, with Holm control over a preregistered, roster-derived family. Both outcomes publish: "no confidence signal helps on strong modern ASR" would be as informative as the confirmation we in fact observe. Beating random deferral is a *floor* — a necessary check that a score is not pure noise, not a claim of optimality against a calibrated or learned selector; we add temperature-calibrated (s_5) and learned-regressor (s_6) comparators in section IV-G.

Novelty positioning (stated verbatim, because it is load-bearing). Conformal prediction for ASR is not new: Ernez *et al.* [11] control WER over sentence *prediction sets* and flag uncertain words with inductive conformal prediction on LibriSpeech/wav2vec 2.0. Their object is a set of hypotheses per utterance; ours is an accept/defer *decision* per utterance with a population-level bound on the accepted set's risk and an explicit coverage/answered-rate trade-off. We claim novelty only for (i) the audit of ASR confidence against an analytic random-deferral excess-AURC null (the selective-classification debunking lens of Traub *et al.* [12], ported to sequence output); (ii) a certified accept/defer gate on accepted-set CER for ASR; and (iii) the released tool. We do not claim "conformal ASR." We also position against the concurrent generic machinery for non-monotone risk control [13] and adaptive selective certificates [14], and against the 2025–26 selective-risk-control line most adjacent to ours: Xu *et al.* [15] and Bai and Jin [16] develop general selective/conformal risk control at the abstract-estimand level; Jia *et al.* [17] is the nearest speech-domain

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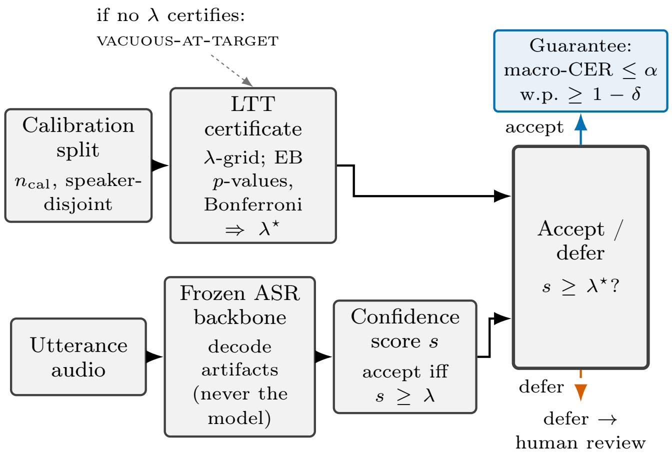

Fig. 1. The `asr-gate` pipeline. Decode artifacts from a frozen ASR backbone (never the model itself) yield a per-utterance confidence score s (e.g. s_1, s_2); a speaker-disjoint calibration split (n_{cal}) drives Learn-Then-Test construction of a threshold $\lambda^{\star}$ (the frozen EB-p-value, Bonferroni-over-grid selective-risk procedure) certifying, with probability at least $1-\delta$, that the accepted set's macro-CER does not exceed α; each utterance is then accepted ($s \geq \lambda^{\star}$) or deferred to a human. If no threshold on the grid certifies, the gate reports VACUOUS-AT-TARGET rather than accepting nothing silently.

instance (coverage-guaranteed *speech-emotion* prediction sets, a classification output, not sequence-CER); and Zhou and Wang [18] audit exactly the bound-tightness and exchangeability strain we concede. A parallel line applies conformal or risk-controlling machinery directly to sequence-generating models rather than to a fixed-vocabulary classifier: Quach *et al.* [19] calibrate a stopping/rejection rule over sampled language-model outputs, Mohri and Hashimoto [20] back off an LM's claims under a conformal factuality guarantee, and Farinhas *et al.* [21] extend conformal risk control to non-exchangeable sequence data (machine translation) — exactly the exchangeability strain we audit empirically via reseeds rather than assume away (section III). These target open-ended text generation or translation; ours targets a fixed decode's accept/defer decision on CER, a narrower and, we argue, more auditable target for a production ASR triage gate. These are methods and diagnoses we could consume, not competitors — our contribution is the ASR-specific instantiation, the audit, and the Mandarin/CER evidence, not new statistical machinery beyond the LTT layer. This is deliberately an *application-and-evaluation-protocol* contribution: the novelty is the ASR-specific instantiation and a reusable evaluation protocol (analytic random-deferral excess-AURC, matched-abstention permutation null, speaker-blocked bootstrap, and a disclosed-deviation discipline), and we calibrate the novelty claim accordingly. Figure 1 summarizes the `asr-gate` pipeline end-to-end.

II. Method: `asr-gate`

`asr-gate` consumes decode artifacts (an N-best list per utterance with sequence log-probabilities and, where available, per-token posteriors) and never the model itself, so the same gate applies to any backbone that emits those artifacts. All statistics come from an in-house reliability-metrics library; the tool "never computes anything the evaluation did not audit."

A. Estimand and the loss

The per-utterance loss is the character error rate $\ell_i = \mathrm{CER}_i = \mathrm{editdist}(\mathrm{hyp}_i, \mathrm{ref}_i)/\mathrm{len}(\mathrm{ref}_i)$, clipped to $[0, 1]$ (insertions can exceed 1; the clip frequency is reported and is 0 on Aishell-1 test). We certify the *macro*-CER (per-utterance mean) of the accepted set. We separately report the field-standard *micro*-CER (total edits over total reference characters) with a speaker-blocked bootstrap confidence interval, but we do not certify it: a micro-CER certificate would need a length-weighted LTT variant, which we flag as future work. This macro/micro distinction is stated once here and carried in every table (table VI). CER is the natural word-free loss for Mandarin, where segmentation is ambiguous; for word-segmented languages the same certificate would instead target WER, and system-independent WER estimation without gold references remains an active research problem [22]. This exact, reference-based ℓ_i also differs from reliability-oriented ASR metrics that score the informativeness/error-aversion trade-off directly, such as RAS [23], which calibrates that trade-off to human preference rather than certifying a finite-sample risk bound.

B. G1: the selective-risk certificate

Let s be a confidence score (higher is more confident) and accept iff $s \geq \lambda$. The selective risk $R(\lambda) = \mathbb{E}[\ell\,\mathbf{1}\{s \geq \lambda\}]/\mathbb{E}[\mathbf{1}\{s \geq \lambda\}]$ is a ratio of two means and is *not monotone* in λ, so plain conformal risk control does not give the C1 statement. We use Learn-then-Test [9]. The key reformulation: because $R(\lambda)$ is a ratio, its Hoeffding–Bentkus p-value is not computed on $R(\lambda)$ directly. The per-λ null $H_0\colon R(\lambda) > \alpha$ is tested through the equivalent bounded-mean statistic $\mathbb{E}[(\ell-\alpha)\mathbf{1}\{s \geq \lambda\}] > 0$, whose summand lies in $[-\alpha, 1-\alpha]$; the two nulls coincide whenever the acceptance probability is positive. We test a grid of λ and select the most permissive rejected threshold $\lambda^{\star}$. With probability at least $1-\delta$ over the calibration draw, the accepted set's macro-CER is at most α. We fix $\delta = 0.1$ and certify targets $\alpha \in \{1, 2, 3, 5\}\%$.

Disclosed pivot: HB $\rightarrow$ EB + Bonferroni (a power fix, with the pilot trace). The preregistered pilot used fixed-sequence testing with Hoeffding–Bentkus (HB) p-values. On the same cached pilot calibration data the two procedures give opposite verdicts at $\alpha = 2\%$, side by side: HB certifies nothing (no threshold rejects, despite an accepted-region CER of 0.95%, well under target), whereas empirical-Bernstein (EB) p-values [24] under a Bonferroni-over-grid correction certify $\alpha = 2\%$ at 86% acceptance. This is a power failure of HB, not a validity failure: HB bounds the deviation using only the fixed $[0, 1]$ range of the per-λ statistic, while EB adapts to its (tiny) sample variance, and for bounded means EB is *a priori* at least as powerful. Both procedures are finite-sample valid, so choosing the more powerful valid one is a power upgrade, not a type-I-error risk — even though the choice was informed by pilot data. We therefore froze the Bonferroni-over-grid procedure with EB p-values before touching the test set, and we disclose both outcomes here (HB the conservative preregistered bound, EB the adopted one) — the deviation is outcome-visible either way — rather than presenting only the final procedure.

C. G2 and the audit

The tool also implements a secondary per-utterance split-conformal upper bound (G2), which we do not evaluate separately here (future work), and a *duration-tercile Mondrian* stratification of the gate thresholds. The reported certificate throughout this paper is the G1 selective-risk certificate; its conditional coverage across the Mondrian duration terciles is reported in section IV-B.

The audit asks whether a score beats honest random deferral. For each score we compute its risk–coverage curve, its AURC, and the analytic random-deferral AURC (closed-form, the risk of accepting a uniformly random subset at each coverage); the excess-AURC is their difference. Significance is a matched-abstention permutation test evaluated at a single operating point — the score's median accept/defer split (the top $\lceil n/2 \rceil$ utterances by score accepted). The test statistic is the accepted-set mean CER; the null re-draws which utterances are deferred while holding the abstention count fixed (within strata when present), preserving the answered rate, and the one-sided p-value is the fraction of $n_{\mathrm{perm}} = 2000$ permutations whose accepted-set mean CER is at most the observed value (floor $1/(n_{\mathrm{perm}}+1) \approx 5.0\times10^{-4}$). This median-split permutation p is a single-operating-point companion to the curve-level excess-AURC statistic, not a re-test of the same quantity. Family-wise error across the score roster is controlled with the Holm procedure [25].

The multiplicity family is roster-derived, and m is recomputed from what actually ran (preregistered). The design specified $m = 10$ (5 scores $\times$ 2 backbones). The preregistered recomputation rule states that "the audit family m is computed from the realized roster after decode gates pass — one test per (auditable score, backbone, condition) actually present with $\geq 99\%$ non-null values." In the main run only Paraformer's s_1 (length-normalized log-posterior) and s_2 (weakest link) survived — Paraformer exposes no N-best margin (s_3) or full posteriors (s_4), and the standalone audit never fits the tune-side temperature for s_5 — so the realized family fell to $m = 2$. In the expansion the realized family is $m = 12$: $\{s_1, s_2\}$ on the six (backbone $\times$ condition) cells that cleared the non-null gate (section IV). The exact m is recorded in each audit result file and reported with the same denominator everywhere; no m is asserted in advance.

D. Honest-uncertainty rules

Out-of-stratum utterances (a stratum with < 200 calibration utterances) are deferred unconditionally. A cheap two-stage detector routes non-Mandarin input to a distinct OOD-REFUSE state (a validity statement, not an uncertainty estimate): on Aishell-1 test this fires on 3 utterances on every reseed, independent of the gate. A domain fingerprint stored in the gate warns when an incoming batch's score distribution shifts beyond a preregistered KS distance (warning only, never silent recalibration). The gate's calibrate and audit modes refuse to run without references; its apply mode runs reference-free by design.

III. Experimental Setup

Data. Aishell-1 [26] is ∼178 h of read Mandarin (16 kHz). We use the official test split (20 speakers, 7,176 utterances, all transcribed, 0 empty), and carve the 40-speaker dev split into speaker-disjoint calibration (20) and tune (20) sets. The cross-corpus arm uses the THCHS-30 test split [27] ($n =$ 2,495 utterances — the full official test split, the ten D-prefix test speakers — from the corrected re-decode; the earlier $n =$ 1,339 decode drew from a *mislabeled mirror split* that was mostly train-speaker audio, not the test split, disclosed below and in table II). The noise arm mixes ESC-50 environmental audio [28] additively onto Aishell-1 test at 5/15/25 dB SNR — a disclosed substitution for the design's MUSAN, which was unreachable at usable bandwidth; the mixer is source-agnostic and the SNR stratification is unchanged. A cross-lingual English arm decodes LibriSpeech [29] test-clean and test-other with Whisper large-v3 and two wav2vec 2.0 checkpoints [30], carving test-clean speaker-disjointly (seed 0) into a calibration split ($n =$ 1,352; fit 739 + cal 613) and a held-in eval-clean split ($n =$ 1,268), and applying the gate to test-other ($n =$ 2,939) as a shifted split (section IV-E). Table I gives the full spec.

Backbones. We attempted three released checkpoints, zero-shot (no training): Paraformer [31], [32] (the primary certified backbone), Whisper large-v3 [33] (zero-shot Mandarin, an out-of-domain stressor), and a ModelScope Conformer-Aishell checkpoint (replacing the design's WeNet backbone [34], which had no installable wheel for the run environment). The Conformer checkpoint emits tokens and text but *no per-token posteriors*, so s_1, s_2 are uncomputable; per the honest-uncertainty rule it is recorded as SKIPPED-DEGRADED and excluded from the audit and certificate — never imputed. This is why the realized roster spans Paraformer and Whisper only.

Reseeds instead of training seeds. No training occurs, so we resample the dev→cal/tune carve at the speaker level 20 times (seeds 0–19), decoding once and recalibrating on cached scores. The empirical violation rate over these 20 reseeds is the primary validity evidence for C1.

Exchangeability candor (load-bearing). Utterances within a speaker are not independent; speaker is the natural block. Cal/tune/test are speaker-disjoint by construction, and all reported CIs are speaker-blocked bootstraps (block = speaker; ∼40 dev / 20 test blocks). The LTT/G2 guarantees assume utterance-level exchangeability between calibration and test, which speaker-level dependence and cal-vs-test speaker disjointness both strain; the 20-reseed violation rate is our direct empirical check on that assumption, and the 20-block test-side bootstrap gives deliberately wide intervals rather than false precision. We state this in the body, not an appendix.

Disclosed protocol deviations are collected in table II: the WeNet→Conformer(degraded) roster change, the HB→EB+Bonferroni pivot, the ESC-50-for-MUSAN and THCHS-mirror substitutions, and the roster-derived m (10 → 2 main, 12 expansion). Data-integrity carry-forward: 17 zero-frame wavs (2 dev / 15 train / 0 test) skipped and logged; 4/7,176 test utterances (0.056%) with degraded token-logp extraction, excluded-and-counted per score. A separate data-integrity anomaly surfaced in the cross-corpus arm, and its accurate mechanism is a *wrong evaluation pool*, not a truncation: the original THCHS-30 decode (n = 1,339, 59 speakers, zero duplicate utterances) drew from a mislabeled mirror split that was 82.7% train-speaker audio (1,107 of 1,339 utterances from train/dev speakers), with only 232 from the genuine test split. We separate the two effects it conflates. (i) *Pool change:* the aligned re-decode on the correct full official test (n = 2,495, ten D-prefix speakers) reports full-set macro-CER 9.93%, versus 15.77% on the contaminated 1,339-utterance mirror pool. (ii) *Like-for-like:* on the 232 utterances that are genuine test utterances in both decodes, macro-CER falls from 16.27% to 9.72% (6.55 pp), isolating the decode fix from the pool change. We disclose the wrong-pool contamination and report the corrected test-split number for the Whisper cross-corpus audit cell (tables V and VI) rather than silently swapping it. The backbone-landscape THCHS-30 cells (table IX, all three backbones) were originally computed on the same 1,339-utterance mirror pool; they have since been re-decoded on the corrected n=2,495 official test (2026-07-16, table II), so table IX and the Whisper audit cell now both report the official-pool numbers.

IV. Results

Every number in this section is read from a single frozen result file, computed only from the archived result artifacts; every figure regenerates from it.

A. The certificate holds on clean Aishell-1 (C1)

Figure 2 shows the s_1 risk–coverage curve on test: the gate tracks well below the analytic random-deferral line and approaches the oracle. Across 20 speaker-level reseeds at $\alpha = 2\%$, $\delta = 0.1$, the certificate holds with 0/20 violations, acceptance fraction 85.7–96.2% (mean 90.7%), and accepted-set macro-CER 1.00–1.53% (mean 1.21%; accepted-set micro-CER 1.02–1.54%, mean 1.23%) — against a full-set macro-CER of 1.98%, so the gate cuts the error rate by roughly 40% (1.98%→1.21% mean) on the accepted ∼91% of traffic. Figure 3 gives the certified frontier: $\alpha = 1\%$ is VACUOUS-AT-TARGET (dev macro-CER already exceeds 1%, so no threshold certifies and 0% is accepted — reported, not silently dropped); $\alpha = 3\%$ and $\alpha = 5\%$ both certify at 99.8% coverage (a grid-floor saturation: the most permissive candidate already satisfies both budgets). The inset shows the 0/20 violation count against the $\delta = 0.1$ line.

The certificate's information on clean data is the conditional accepted-set quality and the binding regimes, not the raw violation count (base-rate disclosure). The full-set macro-CER on clean Paraformer is 1.98%, just under the 2% target, so accept-all already meets an $\alpha = 2\%$ budget and the $\alpha \in \{3, 5\}\%$ certificates saturate at accept-all; what the $\alpha = 2\%$ certificate adds is that it does not accept all — it withholds the low-confidence tail, certifying a restricted 85.7–96.2% set at 1.00–1.53% accepted-set macro-CER. Below the base rate the target genuinely binds, and we map that regime post-hoc by replicating the frozen reseed protocol with only α changed: $\alpha = 1.5\%$ is honestly vacuous (0/20 certified at the frozen $n_{\mathrm{cal}} \approx$ 3,567), while $\alpha = 1.9\%$ — below the 1.98% base rate, so accept-all would fail — certifies in all 20 reseeds at 87.8% mean acceptance and 1.09% accepted-set macro-CER, with 0 violations across the whole 1.5–2.0% band. The certificate therefore does non-trivial work precisely where a budget below the error floor makes accept-all inadmissible.

The $\alpha = 1.5\%$ floor is reachable with more calibration budget (post-hoc). That $\alpha = 1.5\%$ vacuity is itself a calibration-budget artifact, not a property of the bound — the same effect

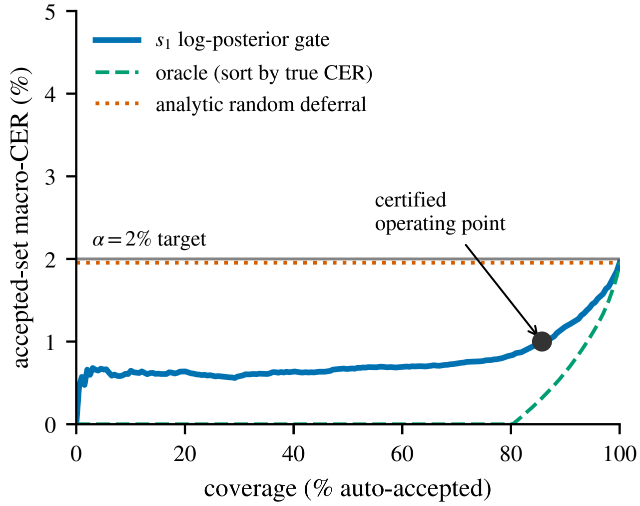

Fig. 2. Risk–coverage on Aishell-1 test (Paraformer, s_1): the confidence gate (blue) tracks well below the analytic random-deferral line (dotted) and approaches the oracle (dashed). The marked operating point is reference reseed 0 (the minimum-acceptance reseed): it auto-accepts 85.7% of utterances at 1.00% macro-CER, under the α=2% target; the mean over 20 reseeds is 90.7% acceptance. (The dotted random-deferral asymptote is the s_1-audit-subset macro-CER, 1.95% over the n=7,172 utterances with a non-null s_1; the 1.98% full-set figure quoted elsewhere is over all n=7,176 utterances — hence the 1.95 vs 1.98% hair.)

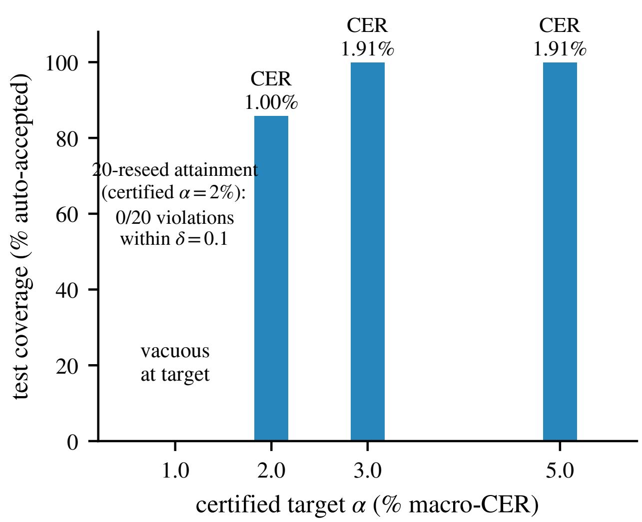

Fig. 3. Certified frontier (Paraformer test), at reference reseed 0 (the minimum-acceptance reseed; the 20-reseed mean acceptance at α=2% is 90.7%). α=1% is VACUOUS-AT-TARGET (0% acceptance); α=2% certifies at 85.7% coverage; α=3% and 5% saturate at 99.8%. The annotation over the vacuous α=1% column reports the α=2% certificate's reseed attainment: 0/20 violations across the 20 reseeds, within the δ=0.1 budget.

TABLE I

DATASET AND SPLIT SPECIFICATION. EVERY BACKBONE GATE CALIBRATES ON A SPEAKER-DISJOINT CARVE OF THE AISHELL-1 DEV SPLIT; EACH TEST SPLIT IS THE OFFICIAL SPLIT, TOUCHED ONCE. THCHS-30, MAGICDATA, AND THE ESC-50 NOISE MIXES ARE APPLY/AUDIT-ONLY (CALIBRATION ALWAYS STAYS ON AISHELL DEV, SO THE MANDARIN CROSS-CORPUS CELLS ARE TRANSFER CELLS). THE THREE LANDSCAPE BACKBONES (PARAFORMER, BELLE, ZIPFORMER) EACH DECODE THE THREE MANDARIN TEST SPLITS IN THE CERTIFIED LANDSCAPE.

Corpus	Role	Utterances	Notes
Aishell-1 dev	cal (20 spk) + tune (20 spk)	14,326	speaker-disjoint carve
Aishell-1 test	certify + audit (3 backbones)	7,176	official, touched once
Aishell-1 test + ESC-50	noise robustness (Para.)	7,176	additive 5/15/25 dB SNR
THCHS-30 test	cross-corpus audit cell (Whisper) + transfer landscape (3 backbones)	2,495	full official test (ten D-prefix spk); corrected re-decode (table II)
MagicData test	cross-corpus transfer (3 backbones)	4,094	CC BY-NC-ND; speaker-disjoint $\sim$4k cap (§2); §D1 substitution
LibriSpeech test-clean	English cal (739+613) + eval-clean	2,620	English arm; speaker-disjoint carve
LibriSpeech test-other	English cross-lingual apply	2,939	English arm; shifted split

TABLE II

DISCLOSED PROTOCOL DEVIATIONS FROM THE DESIGN AND FREEZE NOTE. EACH WAS FIXED BEFORE THE TEST SET WAS TOUCHED, EXCEPT THE ROSTER-DERIVED m (COMPUTED FROM REALIZED DECODES BY PREREGISTERED RULE) AND THE MAGICDATA EVAL CAP (APPLIED POST-HOC TO CACHED SCORES, BY THE RULE FROZEN AHEAD OF THE DECODE).

Deviation	Disposition
WeNet backbone $\to$ Conformer-Aishell	no installable wheel; Conformer then SKIPPED-DEGRADED
Conformer-Aishell in the audit	no per-token posteriors $\Rightarrow s_1, s_2$ null $\Rightarrow$ excluded
Hoeffding–Bentkus $\to$ EB + Bonferroni	pilot power failure (certified nothing at α=2%); disclosed
MUSAN noise $\to$ ESC-50	MUSAN unreachable at bandwidth; source-agnostic mixer
THCHS-30 openslr $\to$ parquet mirror	delivery-route substitution; corrected pool is the full official test (n=2,495, ten D-prefix speakers)
THCHS-30 original decode was the wrong pool	the n=1,339 mirror split was 82.7% train-speaker audio (59 spk, 0 dup), only 232 genuine test utts; corrected to the official test, macro-CER 15.77$\to$9.93% (232-utt like-for-like 16.27$\to$9.72%)
THCHS-30 landscape cells re-decoded on the official test (done 2026-07-16)	the backbone-landscape THCHS-30 cells (table IX, all three backbones) were originally computed on the n=1,339 mislabeled mirror pool; they were re-decoded on the corrected n=2,495 official test (ten D-prefix speakers) on 2026-07-16, and table IX now reports the official-pool numbers. The historical mirror-pool cells are retained frozen but superseded; both THCHS-30 arms (this landscape and the Whisper audit cell) are therefore on the same official test
aidatatang (SLR62) $\to$ MagicData (SLR68)	§D1: SLR62 withdrawn upstream on launch day (two networks); MagicData was the amendment's frozen optional 4th corpus, substituted with no other frozen change
MagicData eval cap (post-hoc)	box decoded the full $\sim$24k test; the frozen §2 speaker-disjoint $\sim$4k cap (9 spk, n=4,094) was applied post-hoc to the cached scores at eval/audit and disclosed
Holm family size m	roster-derived: design 10, main run 2, expansion 12; landscape audit cells recompute m (2 where s_1, s_2 present)
Gender Mondrian axis	dropped: decode records carry no gender label (future work)

the English arm exhibits (section IV-E). A post-hoc calibration-size sweep makes this concrete on the Mandarin arm, with a cleaner design than the English one: we hold the *fixed official test* as eval (calibrating on the speaker-disjoint dev pool, so there is no test-partition calibration caveat) and vary only n_{cal}. The flat Learn-then-Test bound (identical machinery to the English sweep) certifies $\alpha = 1.5\%$ non-vacuously once n_{cal} reaches $\approx 5{,}000$, accepting 83.2% of the fixed test at 0.91% accepted-set macro-CER with 0 violations, stable in 4 of 5 dev-pool re-draws ($n_{\mathrm{cal}} = 7{,}000$ in the fifth). The crossover budget is monotone in the target (α=1.5%$\to$5,000, 1.7% $\to$ 3,567, 1.9% $\to$ 3,000, 5% $\to$ 1,000), qualitatively consistent with the $1/\sqrt{n_{\mathrm{cal}}}$ bound slack. The effect survives the frozen certificate machinery: the main run's calibration gate (duration-tercile Mondrian) on the full dev pool ($n_{\mathrm{cal}} \approx$ 7,184 after the fit/conformalize split), applied to the same fixed test, certifies $\alpha = 1.5\%$ at 90.6% acceptance and 1.20% accepted-set macro-CER, 0 violations — versus 0/20 certified at the frozen $\approx$ 3,567-carve. With $\approx$ 5,000 speaker-disjoint dev calibration utterances the certified frontier therefore tightens from $\alpha = 2\%$ to a binding $\alpha = 1.5\%$ on the same test data, for 0 additional GPU-hours.

Interpreting 0/20 against $\delta = 0.1$ (a correlated diagnostic, read for conservatism not calibration). We are explicit about what this $0/20$ does and does not establish. The 20 reseeds resample only the calibration/tune carve from a fixed 40-speaker dev pool and apply each resulting gate to the same fixed test set, so they are 20 strongly correlated draws, not 20 independent $\mathrm{Bernoulli}(\delta)$ trials; we therefore do not read $0/20$ as a $\pm$ error bar on the violation probability, and it does not certify validity on a fresh test set. What it does show is that the accepted-set risk stays under α across the calibration randomness we can resample here, with margin — an empirically conservative realized rate consistent with the deliberate conservatism of the LTT bound. A valid, finite-sample-honest procedure is expected to under-shoot δ; we would worry only about the reverse (systematic excess), which we do not observe.

Adding the speaker-exchangeability axis (post-hoc). The 20 reseeds vary only the calibration draw; the eval speaker population is fixed, so they cannot probe speaker-level variability in what is evaluated. We add that axis directly: repartitioning the full 60-speaker Aishell-1 pool (dev 40 + test 20) into a 20-speaker calibration cohort and a *disjoint* 20-speaker eval cohort,

$R = 20$ times, and rerunning calibrate+apply at $\alpha = 2\%$, gives $0/20$ violations with the eval speakers themselves changing every partition, at 86.7–95.0% acceptance and 1.00–1.38% accepted-set macro-CER (n_{cal} held at the frozen $\approx 3{,}567$ scale) — ranges essentially indistinguishable from the calibration-only reseed check. The reseed check thus probes calibration randomness and the speaker-partition check probes speaker exchangeability; both pass. This partition check necessarily calibrates on partitions of the official test split, so we report it as a disclosed post-hoc validity probe, not a headline; the main-run numbers still calibrate on dev only.

B. Conditional coverage across duration terciles

The $0/20$ above is a marginal statement. Because the deployed gate is Mondrian in audio duration — its threshold is set per duration tercile, with edges frozen on each reseed's calibration split — we can ask whether the certificate also holds within each tercile. Table III reports, per tercile at $\alpha = 2\%$, the acceptance rate, the accepted-set macro-CER, and the violation count over the 20 reseeds. The certificate holds conditionally: 0/20 violations in every tercile, and every tercile's speaker-blocked bootstrap 95% CI (reference reseed 0) sits below 1.4%, comfortably under the 2% target. The one visible heterogeneity is a mild, expected difficulty gradient: long utterances (dur2) run about 0.3 pp higher in accepted-set macro-CER (1.36% vs 1.06–1.12%) and about 2 pp lower in acceptance (89.8% vs 91.4–91.8%) than short and medium ones, which the Mondrian thresholds absorb by setting a tighter score threshold on the long tercile (reference reseed 0: 0.111 for dur2 vs 0.150 for dur0). This is a difficulty gradient the per-stratum design anticipates, not a failure of control.

C. The certificate degrades gracefully under noise

Figure 4 applies the *clean-calibrated* gate to noisy test audio. At 25 dB and 15 dB SNR the certificate holds with $0/20$ violations (accepted-set macro-CER 1.22% and 1.32%, coverage 90.3% and 89.4%); at 5 dB it shows $1/20$ violations (accepted-set macro-CER 1.62%, up to 2.16% on the single violating reseed, coverage 82.2%). One violation in 20 is a realized rate of 0.05, still within the $\delta = 0.1$ budget: the gate calibrated on clean speech transfers to moderate noise and begins — honestly — to strain only at the most severe 5 dB condition (the largest nominal SNR drop, where the distribution shift the domain fingerprint is meant to flag is plausibly largest).

D. Cross-corpus: the audit transfers where the certificate cannot

The Whisper zero-shot Mandarin cells are a *deliberately weak-backbone stressor*: Whisper large-v3 is not tuned for Mandarin, and its full-set macro-CER is 28.1% on Aishell and 9.93% on THCHS-30 (corrected re-decode, section III) — far above any certified target. They are included not to certify but to exercise the gate's vacuity behaviour and to supply audit-informative cells where the certificate cannot hold. Consequently the Whisper gate is vacuous at every $\alpha \in \{1, 2, 3, 5\}\%$ (0% acceptance): no low-CER accepted region exists to certify (fig. 5 shows the Whisper curves plateauing well above the 2% ceiling while Paraformer dips below it). Yet the audit still fires on Whisper (below), cleanly separating the two claims: confidence can carry selective skill on a backbone too weak to support a low-CER certificate. This is the design's preregistered vacuity outcome, and it is verdict-symmetric — a weak backbone that cannot certify a strong target is a finding, not a failure.

E. Cross-lingual: a non-vacuous English certificate at $\alpha = 5\%$

To test whether the certificate transfers beyond Mandarin, we run an English arm on LibriSpeech with three strong English backbones (Whisper large-v3 and two wav2vec 2.0 checkpoints, base and the larger LV60k-Self self-trained model), calibrating on a speaker-disjoint half of test-clean and applying the gate to the held-in eval-clean split and to the harder test-other split (section III). Table IV reports the outcome at $\alpha = 5\%$. The certificate is non-vacuous: it certifies at $\geq 97.99\%$ acceptance with accepted-set macro-CER 0.61–3.90%, comfortably under the 5% target, on all three backbones and both splits (certified thresholds $\lambda^\star = -0.0959$ for Whisper, -0.1989 for wav2vec 2.0 base, and -0.1498 for wav2vec 2.0 large). On test-other the domain fingerprint fires a shift warning (KS distance 0.176 for Whisper, 0.280 for wav2vec 2.0 base, 0.247 for wav2vec 2.0 large, all above the 0.15 warn threshold), yet the $\alpha = 5\%$ bound is still respected — the gate flags the shift and holds. At the frozen 613-utterance calibration carve the tighter targets are vacuous: at $\alpha \in \{1, 2, 3\}\%$ every cell accepts 0%, as no threshold clears the distribution-free bound on that carve even on these strong backbones. This sub-5% vacuity is a calibration-budget property, not an intrinsic property of the bound. The Learn-then-Test empirical-Bernstein bound slack shrinks with the calibration sample size, so the same held-out test-clean utterances certify tighter targets given more calibration data. A post-hoc calibration-size sweep — holding a fixed speaker-disjoint eval set and varying only n_{cal}, stable across five eval/cal re-draws — makes this concrete: $\alpha = 3\%$ certifies non-vacuously once n_{cal} reaches $\approx 1{,}500$ on all three backbones ($\approx 1{,}000$ on the strongest, wav2vec 2.0 large, which also has the lowest accepted-set error — eval-clean CER 0.61%, test-other 1.70%), and $\alpha = 2\%$ certifies at $n_{\mathrm{cal}} = 1{,}500$ on wav2vec 2.0 large (accepting 96.7% of eval-clean at 0.42% accepted-set macro-CER, a non-vacuous certificate), while Whisper and wav2vec 2.0 base do not reach $\alpha = 2\%$ within the $\approx 2{,}000$-utterance test-clean calibration pool. We report this as practitioner guidance — what calibration data buys — rather than a limitation of the bound: the English arm yields a second language and three backbones under a non-vacuous $\alpha = 5\%$ certificate on the frozen carve, with $\alpha = 3\%$ reachable at a realistic calibration budget.

F. The audit: confidence beats random deferral in all 12 cells (C2)

Figure 6 and table V give the realized $m = 12$ family: $\{s_1, s_2\}$ on Paraformer {clean, +noise 5/15/25 dB} and Whisper {Aishell clean, THCHS-30}. We read the effect sizes, not the p-values, as the primary evidence. With $n \approx 7{,}176$ every

TABLE III
CONDITIONAL COVERAGE ACROSS DURATION TERCILES (PARAFORMER, AISHELL-1 TEST, $\alpha = 2\%$, 20 RESEEDS). TERCILE EDGES ARE FROZEN ON EACH RESEED'S CALIBRATION SPLIT AND APPLIED TO TEST, SO TERCILE SIZES ARE UNEQUAL AND RESEED-DEPENDENT. THE CERTIFICATE HOLDS WITHIN EVERY TERCILE (0/20 VIOLATIONS). THE 95% CI IS A SPEAKER-BLOCKED BOOTSTRAP ON THE REFERENCE RESEED 0 ACCEPTED-SET MACRO-CER (2,000 RESAMPLES, BLOCK = SPEAKER, PERCENTILE), MATCHING THE PAPER'S CI METHODOLOGY. VALUES FROM THE FROZEN TERCILE-ANALYSIS RESULT FILE.

Tercile	Acceptance % (mean [min, max])	Accepted-set macro-CER % (mean [min, max])	Violations	Reseed-0 95% CI
dur0 / short	91.8 [86.8, 96.3]	1.12 [0.92, 1.40]	0/20	$[0.77, 1.09]$
dur1 / medium	91.4 [87.0, 96.2]	1.06 [0.86, 1.33]	0/20	$[0.72, 0.97]$
dur2 / long	89.8 [84.1, 96.2]	1.36 [1.16, 1.75]	0/20	$[1.01, 1.32]$

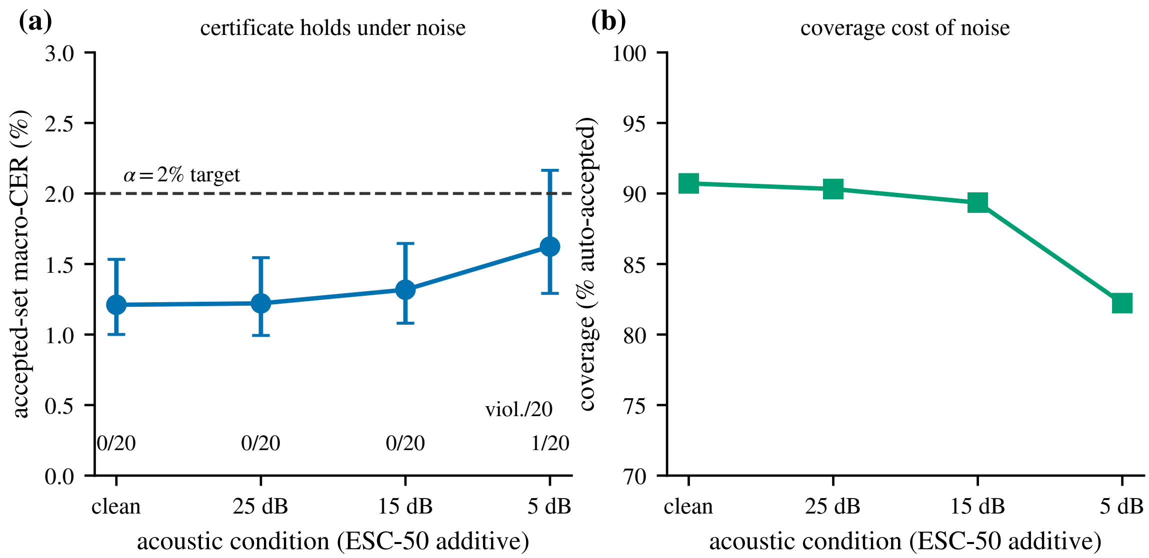

Fig. 4. Noise robustness: the clean-calibrated Paraformer gate applied to ESC-50 additive noise. (a) Accepted-set macro-CER (error bars over 20 reseeds) stays under the α=2% target with 0/20 violations at 25 and 15 dB and 1/20 at 5 dB (within the δ=0.1 budget). (b) Coverage cost of noise.

TABLE IV
CROSS-LINGUAL ENGLISH CERTIFICATE (LIBRISPEECH, LTT, $\alpha = 5\%$, $\delta = 0.1$). CALIBRATED ON A SPEAKER-DISJOINT CARVE OF TEST-CLEAN (n = 1,352; FIT 739 + CAL 613) AND APPLIED TO EVAL-CLEAN (n = 1,268) AND TEST-OTHER (n = 2,939). CERTIFIED THRESHOLDS $\lambda^\star = -0.0959$ (WHISPER), -0.1989 (WAV2VEC 2.0 BASE), -0.1498 (WAV2VEC 2.0 LARGE). "KS SHIFT" IS THE DOMAIN-FINGERPRINT KS DISTANCE TO THE CALIBRATION SCORE DISTRIBUTION (WARN THRESHOLD 0.15); TEST-OTHER EXCEEDS IT, YET THE $\alpha = 5\%$ BOUND IS RESPECTED ON EVERY CELL. ALL CELLS ARE VACUOUS (0% ACCEPTANCE) AT $\alpha \in \{1, 2, 3\}\%$. 20 WHISPER UTTERANCES WERE SKIPPED AT DECODE (11 TEST-CLEAN, 9 TEST-OTHER); BOTH WAV2VEC 2.0 CHECKPOINTS DECODED ALL UTTERANCES. VALUES FROM THE FROZEN ENGLISH-ARM RESULT FILES.

Backbone	Split	Acceptance %	Accepted-set macro-CER %	KS shift
Whisper large-v3	eval-clean	99.45	1.36	—
Whisper large-v3	test-other	97.99	2.33	0.176
wav2vec 2.0 base	eval-clean	100.0	1.23	—
wav2vec 2.0 base	test-other	99.73	3.90	0.280
wav2vec 2.0 large	eval-clean	100.0	0.61	—
wav2vec 2.0 large	test-other	99.63	1.70	0.247

one of the twelve matched-abstention permutation p-values sits at the resolution floor (5.0×10^{-4} at $n_{\text{perm}} = 2000$), so they are all equal and the Holm correction has nothing to discriminate ($p_{\text{Holm}} = 0.006$ throughout); reporting them establishes only that no cell is a null result. The informative quantities are the excess-AURC magnitudes and their uncertainty. Each cell's excess area over analytic random deferral ranges from 0.0118 (Paraformer clean s_1) to 0.0513 (Whisper Aishell clean s_1, the largest in the family, consistent with a weaker backbone having more separable error mass), and its speaker-blocked bootstrap 95% CI (2,000 resamples, blocked by speaker; computed post-hoc from the frozen scored artifacts and prereg-neutral) *excludes zero in all twelve cells* (table V). These are genuine, non-trivial advantages: even the smallest, Paraformer clean, is 0.0118 with CI $[0.0101, 0.0137]$. The advantage is also not an artifact of the median operating point at which the permutation test is evaluated: the accepted-set mean CER falls below the random-deferral expectation at the 25%, 50%, and 75% coverage points in every one of the twelve cells, consistent with the whole-curve excess-AURC being positive. The family now spans two architectures, three noise levels, and two corpora — unlike the toothless $m = 2$ of the main run — which is why the audit claim is carried by the expansion family. These twelve cells are not twelve independent datasets: the four Paraformer

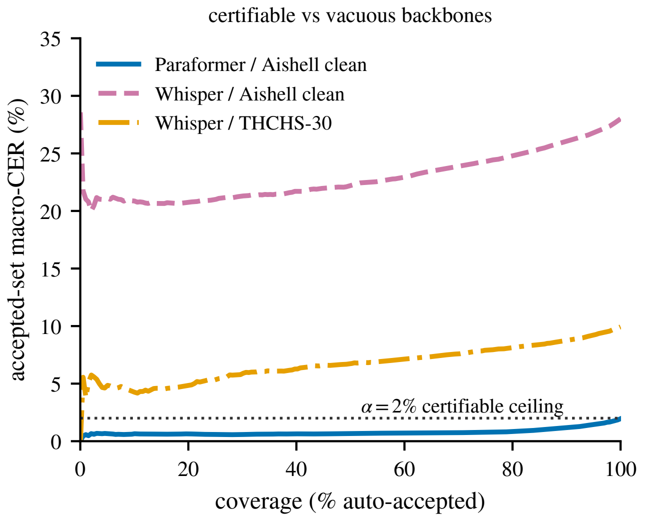

Fig. 5. Why the certificate is vacuous on weak backbones. Paraformer/Aishell dips below the α=2% ceiling (certifiable); Whisper/Aishell and Whisper/THCHS-30 plateau far above it, so no threshold yields a $\leq$ 2% accepted set — even though the audit still finds significant selective skill on these backbones.

conditions re-mix the same Aishell-1 test utterances at different SNRs (clean + 5/15/25 dB), so the cells are dependent; we report per-cell effect sizes and CIs, not a pooled test, and read "twelve cells" as a roster count, not twelve independent corpora. Table VI records the certificate spec and the macro-vs-micro caveat; on Paraformer clean test the full-set macro-CER is 1.98% and micro-CER is 1.94% (speaker-blocked bootstrap CI $[1.72\%, 2.17\%]$, 20 blocks).

G. Comparator scores and speaker-blocked robustness

Two follow-up analyses on Aishell-1 test (Paraformer, $n =$ 7,172; the same 4/7,176 tail rows excluded as above) address the narrowness of the $\{s_1, s_2\}$ family and the utterance-level permutation null. First, fitting the tune-side temperature that the main standalone audit omitted makes s_5 auditable, and we add a learned CER-regressor s_6 as an exploratory comparator. On the primary family $\{s_1, s_2, s_5\}$ (Holm $m = 3$) all three reject at $\alpha = 0.05$ with excess-AURC $0.0118/0.0137/0.0118$ ($p_{\text{Holm}} =$ 0.0015); the exploratory s_6 rejects with excess-AURC 0.0125 ($p_{\text{Holm}} = 0.0005$). The s_5 excess-AURC equals s_1's exactly because temperature scaling is order-preserving and leaves the risk–coverage ranking unchanged — s_5 buys calibration, not ranking. Second, we re-run the matched-abstention permutation null with speaker-level blocking: for all four scores the speaker-blocked p sits at the same resolution floor (5.0×10^{-4}, $n_{\text{perm}} =$ 2000) as the utterance-level null (table VII), so the audit verdict does not depend on within-speaker exchangeability. The s_1, s_2 effect sizes match their Paraformer-clean entries in table V (same substrate).

Named confidence baselines (method-vs-method). Published ASR work reports excess-AURC against *named* confidence estimators [35], so we run the TeLeS-family baselines through the identical audit machinery on the same cells (table VIII); the class-probability and path-probability baselines descend from the word- and utterance-level confidence-estimation lineage [36], [37], [38], and the learned CEM is our implementation of that lineage's logistic confidence-estimation module, not TeLeS's own acoustic-embedding features. The finding is a clarification of scope, not a horse-race win: our two primary scores are two of these standard baselines — min-class-probability equals s_2 exactly, and length-normalized path probability equals $\exp(s_1)$ (a monotone transform, hence an identical excess-AURC) — and the remaining derivable baselines (mean class probability, a learned logistic confidence-estimation module fit on the Aishell-1 dev split) match s_1, s_2 to within ± 0.002 excess-AURC on every certifying cell. No named baseline dominates ours; the contribution is the finite-sample certificate wrapped around these confidence signals, not a novel score. (Tsallis entropy needs the full per-token posterior distribution, absent from the dumps, so it is reported unavailable, not imputed.) The Aishell-1/Belle cell also surfaces on the degraded zipformer transducer (non-certifying everywhere, table IX): its class-probability-derived scores turn *negative* while s_2 and the learned CEM stay positive, a property of that backbone's mis-scaled posteriors, disclosed verdict-symmetrically rather than dropped.

H. The certified landscape: three backbones $\times$ three Mandarin corpora

To move the certificate off a single (backbone, corpus) cell, we preregistered (FREEZE-AMENDMENT 2026-07-13, signed before any new test split was touched) and ran a frozen landscape: three backbones spanning three architectures — Paraformer (B2, NAR-CIF), Belle-whisper-large-v3-zh (B3$'$, attention encoder–decoder), and an offline zipformer transducer [39] (B4, RNN-T) — across three credential-free Mandarin corpora (Aishell-1, THCHS-30, MagicData), on the frozen α-grid $\{1.5, 2, 3, 5, 10\}\%$ with δ=0.1 and 20 speaker-level reseeds per cell. Each backbone's gate is calibrated once on the Aishell-1 dev split and applied to every test split, so the THCHS-30 and MagicData cells are cross-corpus transfer cells. Table IX gives the certified frontier.

Resolution of the single-cell and contamination readings (M1, M2). The frontier now spans six non-vacuous, 0/20-violation certified cells across two backbones $\times$ three corpora, not the single α=2%/Paraformer/Aishell cell — so the "certificate collapses to one cell" reading (M1) no longer holds. (All six legs are on official uncontaminated pools: the two THCHS-30 legs were re-decoded on the corrected n=2,495 official test on 2026-07-16, tables II and IX, joining the two Aishell-1 in-corpus cells and the two MagicData transfer cells, so the single-cell reading breaks on official pools alone.) For contamination (M2): Paraformer-large is an industrial checkpoint whose training data is undisclosed, so Aishell-1 test contamination cannot be excluded from Paraformer alone. *Belle-whisper* answers this by independence, not assertion: it is a different architecture (AED vs. NAR-CIF), from a different organization, with documented training data (Aishell-1/2 train, WenetSpeech, HKUST — test excluded), and it also certifies non-vacuously on the Aishell-1 test split (α=5%, acceptance

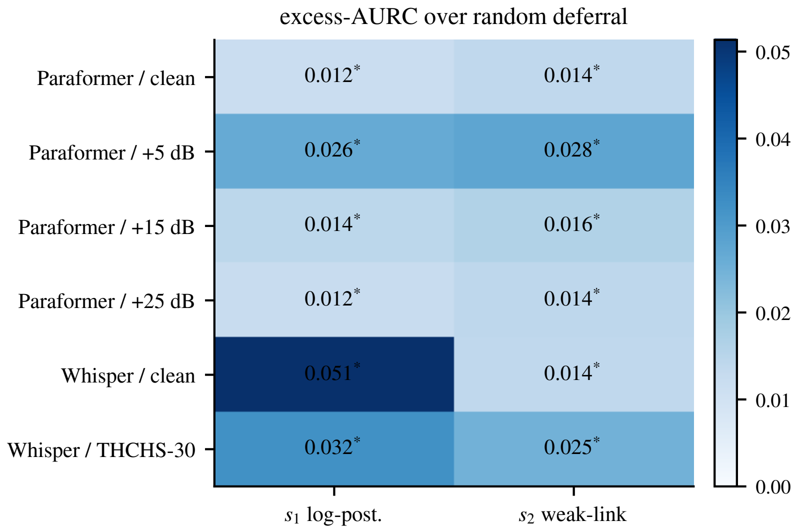

Fig. 6. Realized m=12 audit family: excess-AURC over analytic random deferral for $\{s_1, s_2\}$ across six (backbone × condition) cells. Cell magnitudes are the informative quantity; their speaker-blocked bootstrap 95% CIs all exclude zero (Holm-controlled audit family). The * marks the permutation rejection, which is saturated at the floor for every cell (p_{Holm}=0.006) and hence non-discriminating.

TABLE V
REALIZED AUDIT FAMILY, $m = 12$ (EXPANSION). EXCESS-AURC OVER THE ANALYTIC RANDOM-DEFERRAL BASELINE, WITH A SPEAKER-BLOCKED BOOTSTRAP 95% CI (2,000 RESAMPLES, BLOCK = SPEAKER; COMPUTED POST-HOC FROM THE FROZEN SCORED ARTIFACTS, PREREG-NEUTRAL). THE CI EXCLUDES ZERO IN EVERY CELL. THE MEDIAN-SPLIT MATCHED-ABSTENTION PERMUTATION p-VALUE IS AT THE RESOLUTION FLOOR (5.0×10^{-4}, $n_{\mathrm{perm}} = 2000$) FOR ALL TWELVE CELLS, GIVING $p_{\mathrm{Holm}} = 0.006$ THROUGHOUT — SO THE p-VALUES ARE NON-DISCRIMINATING AND THE EFFECT SIZES AND CIS ARE THE INFORMATIVE QUANTITIES. THE PARAFORMER- AND WHISPER-CLEAN s_2 EXCESS-AURC BOTH ROUND TO 0.0137 BUT ARE DISTINCT (0.01368 AND 0.01370). POINT ESTIMATES AND CIS ARE READ FROM THE FROZEN AUDIT RESULT FILES.

Backbone	Condition	Score	excess-AURC	95% CI (speaker-blocked)
Paraformer	Aishell clean	s_1	0.0118	[0.0101, 0.0137]
Paraformer	Aishell clean	s_2	0.0137	[0.0119, 0.0156]
Paraformer	+noise 5 dB	s_1	0.0264	[0.0195, 0.0348]
Paraformer	+noise 5 dB	s_2	0.0275	[0.0208, 0.0358]
Paraformer	+noise 15 dB	s_1	0.0144	[0.0122, 0.0169]
Paraformer	+noise 15 dB	s_2	0.0161	[0.0138, 0.0187]
Paraformer	+noise 25 dB	s_1	0.0124	[0.0107, 0.0144]
Paraformer	+noise 25 dB	s_2	0.0142	[0.0123, 0.0162]
Whisper	Aishell clean	s_1	0.0513	[0.0462, 0.0567]
Whisper	Aishell clean	s_2	0.0137	[0.0083, 0.0184]
Whisper	THCHS-30	s_1	0.0322	[0.0299, 0.0348]
Whisper	THCHS-30	s_2	0.0248	[0.0228, 0.0266]

TABLE VI
CERTIFICATE SPECIFICATION AND THE MACRO-VS-MICRO CAVEAT. G1 (LTT WITH THE BONFERRONI-OVER-GRID PROCEDURE AND EMPIRICAL-BERNSTEIN p-VALUES) CERTIFIES ACCEPTED-SET MACRO-CER; MICRO-CER IS REPORTED WITH A SPEAKER-BLOCKED BOOTSTRAP CI BUT IS NOT CERTIFIED. CLIP FREQUENCY (CER $>$ 1) IS 0 ON AISHELL-1 TEST. VALUES FROM THE MAIN-RUN AND EXPANSION AUDIT RESULT FILES. THE ACCEPTED-SET (CERTIFIED) ROW'S MICRO-CER IS A DESCRIPTIVE 20-RESEED COMPANION, NOT CERTIFIED.

Backbone / condition	Statistic	macro-CER	micro-CER
Paraformer / Aishell clean (full set)	reported	1.98%	1.94%
Paraformer / +noise 5 dB (full set)	reported	3.80%	3.74%
Paraformer / +noise 15 dB (full set)	reported	2.30%	2.26%
Paraformer / +noise 25 dB (full set)	reported	2.04%	2.00%
Whisper / Aishell clean (full set)	reported	28.06%	26.17%
Whisper / THCHS-30 (full set, fixed re-decode)	reported	9.93%	9.95%
Paraformer / Aishell clean (accepted, α=2%)	**certified**	1.00–1.53%	1.02–1.54%

93.3%, accepted-set macro-CER 4.20%, 0/20). Simultaneous independent memorization of the Aishell-1 test split by two backbones this different is not a credible single explanation, which bounds the contamination card. **Verdict-symmetric caveat.** The frozen roster's fully-open transducer (B4, whose public training data provably excludes the Aishell-1 test) was named as the single strongest contamination answer, but it failed to expose usable posteriors (Aishell full-set CER 45.88%;

TABLE VII

COMPARATOR AUDIT ON AISHELL-1 TEST (PARAFORMER, $n = 7{,}172$). PRIMARY FAMILY $\{s_1, s_2, s_5\}$ UNDER HOLM ($m = 3$); s_6 (LEARNED CER-REGRESSOR) IS AN EXPLORATORY COMPARATOR ($m = 1$). ALL REJECT AT $\alpha = 0.05$. THE LAST COLUMN GIVES THE MEDIAN-SPLIT MATCHED-ABSTENTION PERMUTATION p UNDER UTTERANCE-LEVEL AND SPEAKER-BLOCKED NULLS ($n_{\mathrm{perm}} = 2000$, FLOOR 5.0×10^{-4}): BLOCKING DOES NOT WEAKEN THE VERDICT. s_5'S EXCESS-AURC EQUALS s_1'S BECAUSE TEMPERATURE SCALING IS RANK-PRESERVING. VALUES FROM THE FROZEN COMPARATOR-AUDIT RESULT FILE.

Score	excess-AURC	p_{Holm}	perm. null p (utt. / spk-blocked)
s_1	0.0118	0.0015	5.0×10^{-4} / 5.0×10^{-4}
s_2	0.0137	0.0015	5.0×10^{-4} / 5.0×10^{-4}
s_5	0.0118	0.0015	5.0×10^{-4} / 5.0×10^{-4}
s_6 (expl.)	0.0125	0.0005	5.0×10^{-4} / 5.0×10^{-4}

TABLE VIII

NAMED CONFIDENCE-BASELINE EXCESS-AURC, EXPANDED TO EVERY BACKBONE $\times$ CORPUS CELL THE STANDALONE BASELINE RUN COMPUTED (AISHELL-1, MAGICDATA; THCHS-30 CELLS OMITTED HERE, SEE NOTE BELOW), THROUGH THE SAME AUDIT MACHINERY (n_{perm}=2000). CLASS-PROB-MIN $\equiv s_2$ AND LENGTHNORM-PATH-PROB $\equiv \exp(s_1)$ BY CONSTRUCTION (IDENTICAL RANKINGS); MEAN CLASS PROB AND THE LEARNED CEM ARE THE GENUINELY DISTINCT BASELINES. ON PARAFORMER/BELLE (THE CERTIFYING BACKBONES) ALL COINCIDE WITH OUR SCORES WITHIN ± 0.002; ON THE NON-CERTIFYING ZIPFORMER TRANSDUCER, s_1/CLASS-PROB/LENGTHNORM TURN NEGATIVE WHILE s_2/CEM STAY POSITIVE (VERDICT-SYMMETRIC, CONSISTENT WITH THE LANDSCAPE VAC. VERDICT FOR ZIPFORMER). PARAFORMER/MAGICDATA IS EXCLUDED WHOLESALE (41/4,094, 1.0% OF ROWS LACK A DECODED SCORE — THE FAMILY NEVER COMPUTED, NOT IMPUTED). MAGICDATA CAPPED TO n=4,094 AS IN THE CERTIFIED LANDSCAPE. TSALLIS ENTROPY: UNAVAILABLE (NO FULL POSTERIORS). THCHS-30 CELLS ARE OMITTED FROM THIS TABLE: THE STANDALONE BASELINE RUN (2026-07-15) PREDATES THE 2026-07-16 OFFICIAL-POOL THCHS-30 REDECODE (DISCLOSED PROTOCOL DEVIATIONS) AND ITS THCHS-30 NUMBERS ARE ON THE SUPERSEDED n=1,339 MISLABELED-MIRROR POOL – REPORTING THEM BESIDE THE CORRECTED POOL'S LANDSCAPE NUMBERS WOULD SILENTLY REINTRODUCE THE CONTAMINATION THIS PAPER DISCLOSES AND FIXES ELSEWHERE, SO THEY ARE EXCLUDED RATHER THAN SHOWN STALE.

Backbone	Corpus	macro-CER	s_1	s_2	class-prob (mean)	class-prob (min) $\equiv s_2$	lengthnorm path $\equiv \exp s_1$	learned CEM
Paraformer	Aishell-1	1.98%	0.0118	0.0137	0.0117	0.0137	0.0118	0.0122
Paraformer	MagicData	4.89%	excluded: 41/4,094 (1.0%) rows missing every score, family dropped					
Belle	Aishell-1	5.12%	0.0236	0.0221	0.0237	0.0221	0.0236	0.0199
Belle	MagicData	7.80%	0.0420	0.0372	0.0422	0.0372	0.0420	0.0346
zipformer	Aishell-1	45.88%	−0.0158	0.0578	−0.0226	0.0578	−0.0158	0.1890
zipformer	MagicData	32.21%	−0.0055	0.0567	−0.0102	0.0567	−0.0055	0.1504

TABLE IX

CERTIFIED LANDSCAPE (FROZEN PROTOCOL; 20 RESEEDS/CELL, δ=0.1). "TIGHTEST CERT. α" IS THE SMALLEST GRID α CERTIFYING NON-VACUOUSLY WITH 0/20 RESEED VIOLATIONS; ACCEPTANCE AND ACCEPTED-SET MACRO-CER ARE ITS 20-RESEED MEANS. AISHELL-1 IS IN-CORPUS (OFFICIAL TEST, n=7,176); THCHS-30 (n=2,495, THE CORRECTED FULL OFFICIAL TEST—TEN D-PREFIX TEST SPEAKERS, RE-DECODED 2026-07-16, DISCLOSED AS A PROTOCOL DEVIATION) AND MAGICDATA (SPEAKER-DISJOINT $\sim$4K EVAL CAP, n=4,094) ARE AISHELL-DEV-CALIBRATED TRANSFER CELLS. VAC. = VACUOUS AT EVERY GRID α; FOR THESE ROWS THE ACCEPT AND ACC-CER CELLS ARE — (NO SET IS ACCEPTED, SO BOTH ARE UNDEFINED).

Backbone	Corpus	full-set CER	tightest cert. α	accept	acc-CER
Paraformer (B2)	Aishell-1	1.98%	1.9% †	87.8%	1.09%
Paraformer (B2)	THCHS-30	4.07%	5%	100%	4.06%
Paraformer (B2)	MagicData	4.89%	5%	98.1%	4.12%
Belle (B3′)	Aishell-1	5.12%	5%	93.3%	4.20%
Belle (B3′)	THCHS-30	6.99%	10%	100%	6.99%
Belle (B3′)	MagicData	7.80%	10%	96.9%	6.70%
zipformer (B4)	Aishell-1	45.88%	VAC.	—	—
zipformer (B4)	THCHS-30	82.15%	VAC.	—	—
zipformer (B4)	MagicData	32.21%	VAC.	—	—

its length-normalized-logprob confidence s_1 scores a negative excess-AURC, worse than random deferral), so B4 is vacuous at every cell and is reported non-certifying / degraded-posterior — exactly as the Conformer cell is. The fully-open leg is therefore not delivered, and sub-3% certification on Aishell-1 remains Paraformer-only; the contamination answer rests on Belle's documented-data independence.

Belle's certificate is calibration-budget-robust at $\alpha = 5\%$, but not tighter (post-hoc). Because the contamination answer rests on Belle, we ran a post-hoc calibration-size power curve on it, mirroring the English/Mandarin sweeps' design (fixed official Aishell-1 test as eval; vary only n_{cal} over the speaker-disjoint dev pool; here R=20 budget reseeds per cell, reporting the fraction that certify non-vacuously rather than a single-seed crossover). The reported α=5% floor is cheap to reach: the flat LTT bound (identical machinery to the other sweeps) certifies α=5% non-vacuously in $\geq 90\%$ of reseeds by n_{cal}=2,000 — about $4\times$ smaller than the frozen $\approx 3{,}567$-utterance gate budget already in use — with 0 violations anywhere on the grid. Unlike Paraformer, though, more budget does not keep tightening the frontier: α=3% reaches $\geq 90\%$ of reseeds only at the full 14,326-utterance dev pool (accepting a low 49.9% of the fixed test at 2.51% accepted-set macro-CER), and $\alpha \leq 2\%$ never certifies non-vacuously in any of the 20 reseeds at any tested budget, including the full pool. Belle's contamination-answer certificate is therefore calibration-budget-robust at its reported operating point, but its sub-3% ceiling looks like a backbone/score-separation limit rather than a calibration-budget one, consistent with the sub-3%-is-Paraformer-only statement in section VI.

V. Discussion

The two claims are deliberately separable. C1 is a guarantee whose usefulness is contingent on the backbone: it is informative on Paraformer (error cut by roughly 40% at 91% coverage), robust to moderate noise, and honestly vacuous on a backbone whose floor CER exceeds the target. C2 is a debunking test that survives independently of C1 — it confirms, across a broad realized family, that field-standard ASR confidence is not merely random, which is the precondition any selective deployment implicitly assumes yet is seldom tested directly [12]. Relative to Ernez *et al.* [11] we certify an accept/defer decision's population risk rather than the WER of a prediction set; relative to generic non-monotone risk control [13], [14] we contribute the ASR instantiation, the random-deferral audit, and the Mandarin/CER evidence, not new machinery. The apparatus (analytic random-deferral excess-AURC, matched-abstention null, speaker-blocked bootstrap, conformal gate) transplants from selective classification and off-policy evaluation to sequence-output ASR with only the LTT layer added. Distribution-free risk control has been deployed analogously elsewhere: Fisch *et al.* [40] bound false positives in multi-label conformal sets, Schuster *et al.* [41] calibrate an early-exit stopping rule for autoregressive decoding under a similar Learn-then-Test-style guarantee, and Tibshirani *et al.* [42] extend conformal validity under covariate shift — the same distributional-drift concern our domain fingerprint flags rather than resolves (section II-D).

VI. Limitations

The certificate's tightest ($\leq 3\%$) statement rests on one backbone (Paraformer): the documented-data Belle backbone certifies on Aishell-1 test only at $\alpha \geq 5\%$ (a post-hoc calibration-size power curve shows this floor is calibration-budget-robust — non-vacuous in $\geq 90\%$ of 20 reseeds by n_{cal}=2,000 — but $\alpha \leq 2\%$ never certifies even at the full 14,326-utterance dev pool, so the floor past α=3% looks like a backbone/score limit, not a budget one), the fully-open zipformer failed to expose usable posteriors and certifies nowhere, and the Conformer-Aishell/Whisper cells are degraded-excluded or vacuous (section IV-H). So while the landscape breaks the single-cell reading, the sub-3% regime is still Paraformer-only. Scores s_3, s_4 are structurally null on Paraformer (no N-best margin or full posteriors); s_5 becomes auditable only once the tune-side temperature is fit, and a learned s_6 is reported as an exploratory comparator (section IV-G). The main run's $m = 2$ Holm family is nearly toothless; the multiplicity claim rests on the expansion's $m = 12$. Twenty test speakers give only 20 blocks for the speaker-blocked bootstrap, so those intervals are wide by construction (the audit's permutation verdict, however, is unchanged under speaker-level blocking, section IV-G). The certificate is on macro-CER; micro-CER is reported, not certified. The cross-lingual English certificate is non-vacuous at $\alpha = 5\%$ on the frozen 613-utterance carve and vacuous at $\alpha \leq 3\%$ there; a post-hoc calibration-size sweep shows this sub-5% vacuity is a calibration-budget effect, $\alpha = 3\%$ (and $\alpha = 2\%$ on the strongest backbone) becoming certifiable at a larger calibration budget (section IV-E). The Mandarin cross-corpus certificate that an earlier draft flagged as the highest-value follow-up is now delivered: Paraformer and Belle both certify non-vacuously as transfer cells on MagicData and on THCHS-30 (section IV-H); the two THCHS-30 legs were re-decoded on the corrected n=2,495 official test (table II), so both MagicData and THCHS-30 are uncontaminated cross-corpus transfer legs. All results are on read Mandarin and English plus additive-noise and cross-corpus stress; spontaneous-speech robustness is untested. Broader Mandarin corpora exist beyond our credential-free three — AISHELL-2's industrial-scale 1000-hour read corpus [43] and WenetSpeech's 10000+-hour multi-domain web/podcast collection [44] — both requiring heavier licensing or download infrastructure than the SLR-hosted corpora used here; extending the landscape to spontaneous, multi-domain Mandarin speech is exactly the corpus axis WenetSpeech would add, left for future work.

Data and Code Availability

`asr-gate` and all frozen result files are released in a public GitHub repository and archived under a Zenodo DOI for reproduction (TODO-USER: insert the repository URL and Zenodo DOI before submission); every figure and table regenerates from those files via the released number-registry and figure-generation code, so all reported numbers are reproducible bit-for-bit from the archived artifacts. The noise corpus (ESC-50, CC BY-NC family) is used as data only; no CC-BY-NC code or weights are executed anywhere in the pipeline. For frozen-artifact integrity we keep the original result filenames, which still carry the MUSAN SNR labels even though the realized noise corpus is ESC-50: this is the disclosed MUSAN→ESC-50 substitution (table II), and the filenames are preserved unchanged rather than renamed post hoc.

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